# Abstract

Intelligent reflecting surfaces (IRS) have emerged as a promising alternative to conventional decode-and-forward (DF) relaying for coverage extension in wireless networks. While prior analytical studies have demonstrated the $N^2$ beamforming gain of IRS under deterministic channel assumptions, the performance under realistic stochastic fading remains insufficiently characterised. This paper extends the analytical framework of Björnson et al. (2020) by evaluating IRS and DF relay performance under the 3GPP Spatial Channel Model (SCM) with variable Ricean $K$-factor, pathloss exponent, and scattering cluster count. Using spectral efficiency as the primary metric, the study conducts four parameter sweeps across $\mathbf{1000}$ Monte Carlo realisations in a 3GPP Urban Micro-cell (UMi) deployment at $\mathbf{3\ GHz}$. Results show that IRS with $N=200$ elements outperforms DF relay by $+56.8\%$ to $+64.3\%$ in median spectral efficiency across all channel conditions tested. The 10th percentile (outage) analysis reveals that IRS is $1.8\times$ more sensitive to $K$-factor than DF relay: the IRS outage-to-median ratio improves from $\mathbf{75.7}\%$ ( $K=0$ $\mathbf{dB}$ ) to $98.1\%(K=25\ \mathrm{dB})$, approaching the DF relay stability of $98.9\%$. An extended pathloss exponent sweep ($\alpha_{\mathrm{PL}} \in \{0.5,1.0,2.0,3.0,4.5\}$) demonstrates that IRS achieves its largest advantage in guided-propagation environments ( $\alpha_{\mathrm{PL}} \leq 2.0$ ), where the cascaded effective exponent remains at or below free-space levels. The cluster count sweep ( $L \in \{1,3,6,20\}$ ) shows that multipath diversity steepens the spectral efficiency distribution without shifting the median, providing "free" reliability enhancement. The minimum number of IRS elements required to outperform DF relay, $N_{\min}$, decreases by $\mathbf{11.1}\%$ from $K=0$ $\mathbf{dB}$ to $K=25$ $\mathbf{dB}$, with diminishing returns beyond $K=10$ $\mathbf{dB}$. Design guidelines for IRS deployment are provided, including $K$-dependent sizing formulas, environment suitability criteria, and frequencyband recommendations.